Winter is Coming: When the AI Hype Fades

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By late October 2025, NVIDIA, the shovel maker of the AI gold rush, was valued higher than the GDP of every nation except the United States and China.[1] And it was not alone. OpenAI's valuation jumped from $29 billion in 2023 to a staggering $500 billion in 2025.[2] Is the AI hype reshaping humanity for good, or is it the largest bubble since the Dot-Com Crash, poised to explode?

To answer that question, we first need to define what a bubble is. As Yale economist Robert Shiller puts it, a bubble occurs when the price of something grows very fast, mainly driven by investors' enthusiasm and fear of missing out, with little or no relationship to its fundamentals.[3] The most infamous example would be the Dot-Com Bubble that ballooned during the late 1990s. Investors treated new metrics like website traffic as if they were gold, and poured money into any company with a ".com" suffix, even when its business model is untested.[4]

Let us now examine whether the current AI hype fits the portrait of a bubble. Enthusiasm? Surely. Fear of missing out? Without question. But what about the fundamentals? In other words, will today's AI technology—namely the Large Language Model (LLM)—be as profitable as it claims to be? The sad truth is that it will not.

The first problem is that today's LLMs may be hitting a hard ceiling, whether in their capabilities, data supply, or other resources required to scale them. Yann LeCun, a Turing Award winner, argues that we will not reach Artificial General Intelligence (AGI) simply by scaling up LLMs. It is like trying to reach the moon by climbing progressively taller trees. There is also the issue of diminishing marginal returns. GPT-4, which is ten times larger than GPT-3, was a massive success. Many of us believed GPT-5, which is ten times larger than GPT-4, would deliver true AGI. Yet it turned out to be only an incremental improvement. It was like expecting a spaceship and getting a slightly faster Tesla instead.

As for data supply, Ilya Sutskever, the co-founder of OpenAI, has warned that scaling LLMs is reaching a plateau, and that we are running out of high-quality human text on the internet to feed these models. Adding to the challenge, obtaining new data is becoming increasingly difficult, with lawsuits over data ownership popping up everywhere. Synthetic data generated by AI can help, but it is no silver bullet. LLMs still need to be grounded in authentic human text to function effectively.

More disturbingly, one of the bottlenecks in scaling up LLMs may be energy. According to International Energy Agency, electricity demand from data centers alone will reach 945 TWh by 2030, accounting for 3% of global electricity consumption.[5] To put that in perspective, aluminium industry accounted for only 4% of global electricity consumption in 2023.[6] At the same time, tech giants are continuously falling short of their green promises. Microsoft, for instance, pledged to be carbon negative by 2030, yet its total carbon emissions have risen nearly 30% since 2020 due to data center expansion.[7] This raises two questions. First, can our current energy infrastructure sustain the demands of AI development? Second, can AI contribute to solving global warming, given the irony that its own growth is becoming a major source of emissions? The answer may well be no.

So LLMs are likely just an early result in the pursuit of human-level AI, and possibly even a dead end. There may not be a single day before which AGI does not exist and after which it suddenly does. History reminds us to be wary of the delusion that AGI can be created in just a few years with seemingly sufficient money and a bunch of geniuses. In the early 1980s, a wave of excitement around expert systems prompted Japan to invest heavily in the Fifth Generation Computer Project, which turned out to be mostly a failure.

But even if AGI is created, that raises a second and equally difficult problem: when will it become sufficiently reliable? In the deployment of new technologies, especially AI, reliability is often where progress falters. This challenge is not new. It is the same reason we saw impressive autonomous driving demos a decade ago, yet still do not have Level 5 self-driving cars today. The last few percent of reliability are what make a system truly usable. Beyond that, we must also learn how to integrate the technology into existing infrastructure and ensure that it makes its users more efficient.

Look at what happened with IBM Watson Health. IBM wagered billions on the promise that Watson could learn medicine and then be deployed in every hospital to generate massive revenue. Yet, it turned out to be a complete failure. Clinicians lost trust when its recommendations failed to align with current medical standards.[8] Due to the lack of explainability, Watson functioned like a black box making high-stakes decisions. Eventually, the project was sold for a fraction of its initial investment.

My opponents might argue that today's AI technologies—not just LLMs—are fundamentally more capable than past failures. They continue to innovate at an unprecedented pace, so perhaps the current AI hype is not a bubble at all. Yet this brings us to a third, far more pressing problem: AI companies are generating valuations that their current revenues can not justify. According to Bain & Company's sixth annual Global Technology Report, AI firms would need to make $2 trillion a year in revenue by 2030 just to fund the required computing power.[9] To put that in perspective: in 2024, Alphabet, Amazon, Apple, Meta, Microsoft and NVIDIA combined did not make $2 trillion. In other words, AI is already priced as if it will become the largest money-making engine in the history of capitalism—bigger than every major tech company today combined. But in reality, only 5% of generative AI use cases have delivered profitability.[10]

An even more serious red flag is that AI firms are all being funded with a ton of debt, and a huge part of that debt is hidden through circular financing. To illustrate this, let us suppose that NVIDIA, the chipmaker, decides to invest in OpenAI, saying, "Here's $100 billion, go nuts." OpenAI is like "Cool, we have money," and commits to $100 billion in contracts with Oracle to run ChatGPT. Then Oracle is like "Wow, we're making money." so they buy $100 billion worth of GPUs from NVIDIA. In this way, the money flows from NVIDIA to OpenAI, from OpenAI to Oracle, and then from Oracle right back to NVIDIA, creating the illusion of real revenue. What all of this does is inflating the stock market and making the U.S. economy appear stronger than it really is. Without the AI hype, the economy would likely be in a recession right now.

If you add all this up and look at what some of the biggest investors are doing, you might be worried. Warren Buffett, for example, has sold billions in stock and now sits on the largest cash pile in the world.[11] So the real question is not whether AI is a bubble, but what kind of bubble it might be: a 2001-type crash that dealt a heavy blow to the tech industry, a 2008-type crisis that shook the Western economy, or a 1929-type depression that brought down the whole world's economy.

A 1929-style depression seems unlikely in the short term. The fragile financial system of the 1920s, propped up by the *Dawes Plan* and *Young Plan*, no longer exists. Modern economies benefit from robust banking regulations and macroeconomic tools influenced by the New Deal and Keynesian theory. Similarly, a 2008-type crisis appears less probable. Unlike the pre-2008 financial system, today's markets are not built on excessive debt. Most major AI players are part of larger companies, reducing the risk of a systemic collapse.

Looking back just three decades, the Dot-Com Crash seems like a rehearsal to what is happening today. Many features we see in the current AI hype—concentrated capital, temporary unprofitability and valuations driven by future promise—were also observed in the Dot-Com Bubble. In terms of relative growth, the Dow Jones rocketed 200% from 2019 to 2025, catching up to the 250% climb from 1995 to 2001.[12] In terms of scale, however, the bursting of the current AI bubble could be even more severe than the Dot-Com Crash.

Once it begins to burst, countless AI firms and financial institutions could collapse, slowing down the entire economy. In the worst case, this could escalate into a full-scale financial meltdown. In response, the U.S. and Chinese governments are likely to intervene to rescue these companies at all costs because neither country can afford to lose the AI arms race. Trade protectionism may resurface, and U.S.-China relations could deteriorate once again. At the end of the coming AI winter, new giants will rise from the ashes, just like Google and Amazon did after the Dot-Com Crash, leading to even greater monopolization within the AI industry.

At the end of this essay, I would like to explain why part of me have no desire to witness the birth of AGI, at least for now. Seven years after the stock market crash of 1929, John Maynard Keynes published *The General Theory of Employment, Interest and Money*, where he delivered his most famous idea: "poverty in the midst of plenty."[13] The more advanced a society becomes in production, the worse the problem of under-consumption becomes, so that it must halt production in the cruelest way possible—through unemployment. That is why the United States, the richest nation on Earth, suffered the worst crisis in human history.

So how can we break this cycle? Keynes' solution was not wealth redistribution, but monetary expansion—massive government borrowing and continuous liquidity injection by the central bank to keep the system running. Yet, this excess liquidity only amplifies the boom-and-bust cycle. Each crisis requires more stimulus than the last; the snowball grows larger, heavier, more destructive, until it reaches a scale beyond human comprehension. And when that day comes, what can we do?

As Keynes put it, "In the long run, we are all dead."[14] In other words, the goal is simply to delay the inevitable long enough for the next technological revolution—or the next world war—to reset the system. Even if the current AI hype does reword us with AGI, that will not be the end of this story. This story may have no end at all. The inherent crises of capitalism will continue to hang over humanity like the sword of Damocles. However, if AGI never arrives, Keynes' solution may finally fail, bringing about the collapse of capitalism. By then, human civilization will find a way forward—as it always does.

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